

# Isolation cone centering fix: $|v_z|$ -dependent mis-centering between cluster COG and tower centers, and an ownership-excluded iso variant

PPG12 Analysis Team

April 18, 2026

## Summary

The manual tower-based and topo-cluster isolation computations in `anatreemaker/source/CaloAna24.cc` used `RawClusterUtility::GetPseudorapidity(cluster, vertex)` as the cone axis. That projects the cluster's stored position (`RawCluster::get_position`), which carries the sPHENIX shower-depth correction and sits at  $R_{\text{cluster}} \approx 95$  cm, while the constituents (towers via `RawTowerGeom::get_center_{x,y,z}`, or topo-cluster centroids) live at tower mid-depth  $R_{\text{tow}} \approx 102$  cm. Projecting through a non-zero primary vertex  $v_z$  turns the  $\sim 7$  cm radial gap into a  $|v_z|$ -proportional  $\eta$  offset ( $\Delta\eta \simeq |v_z|(1/R_{\text{cluster}} - 1/R_{\text{tow}})$ ). The offset exceeds  $R = 0.05$  around  $|v_z| \gtrsim 60$  cm, so the cluster's own central tower lands outside a small- $R$  cone even though it is the dominant tower of the cluster itself. Because the stored iso is (sum in cone)  $- \text{cluster\_}E_T$ , this produced the symptom `cluster_iso_005`  $\approx -\text{cluster\_}E_T$  for **100 %** of clusters at  $|v_z| > 80$  cm (and **23.7 %** overall in a Run 24 data sample) — an unphysical value that can only appear if the cone fails to contain the cluster.

The fix replaces the cone axis with the *CoG-tower axis* (geometry lookup of the central tower `showershape[4,5]` projected through  $v_z$ ), and adds a parallel `cluster_iso_excl_*` family that skips cluster-owned EMCal towers by TowerInfo key (identity-based self-exclusion, no scalar  $-\text{cluster\_}E_T$  subtraction). All full-calorimetry iso branches (`cluster_iso_{005, 0075, 02, 03, 04}` and `cluster_iso_excl_*`) now use a common 120 MeV per-tower threshold and the CoG-tower axis. The sPHENIX built-in `RawCluster::get_et_iso` accessor is no longer read; `cluster_iso_{02,03,04}` are computed manually on the same axis and threshold as `cluster_iso_{005, 0075}`.

**Empirical verification.** A re-run of the `anatreemaker` test pipeline on 5 000 events of Run 24 data (196 clusters with  $E_T > 1$  GeV) gives **0.0 %** pathological `cluster_iso_005` across every  $|v_z|$  bin, and 100 % per-cluster monotonicity  $\text{iso}_{005} \leq \text{iso}_{02} \leq \text{iso}_{04}$ .

## 1 Observed symptom

At small cone radii, the ratio `cluster_iso_005 / cluster_` $E_T$  piled up at  $-1$  for a sizable fraction of clusters, meaning the tower sum inside  $R = 0.05$  was vanishing and the stored value was dominated by the  $-\text{cluster\_}E_T$  subtraction. Figure 1 shows the 1D distribution of the ratio in a Run 24 data sample, before and after the fix. The population near  $-1$  (*cone misses cluster*) is cleanly removed.

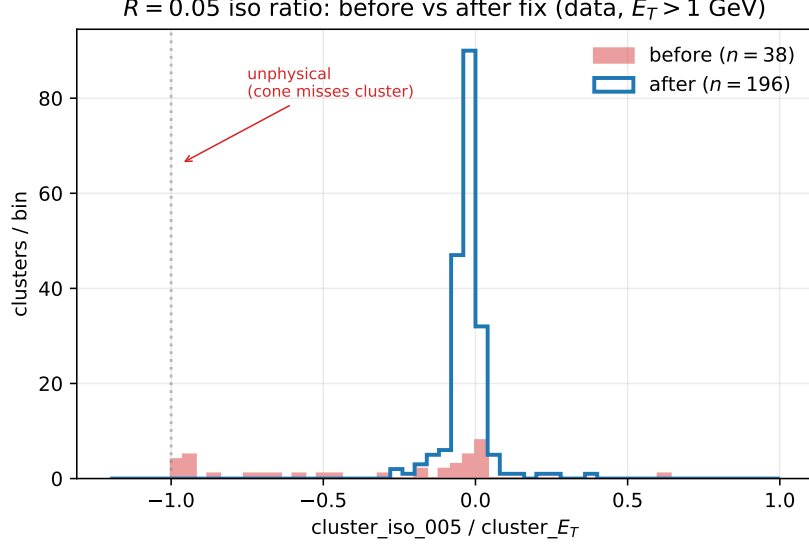


Figure 1: Distribution of `cluster_iso_005` / `cluster_ET` in Run 24 data ( $E_T > 1$  GeV). The pile-up at  $-1$  (before fix, red) is the fingerprint of the cone geometrically missing the cluster itself. After fix (blue) the distribution is physical, with no population near  $-1$ .

### 1.1 $|v_z|$ -correlation identified the depth mismatch

The smoking gun was that the pathology is strongly correlated with the primary vertex position. Figure 2 plots the ratio against  $|v_z|$ : before the fix, clusters at  $|v_z| > 60$  cm drive straight to the unphysical  $-1$  line, while low- $|v_z|$  clusters are healthy. After the fix, the ratio is flat in  $|v_z|$ .

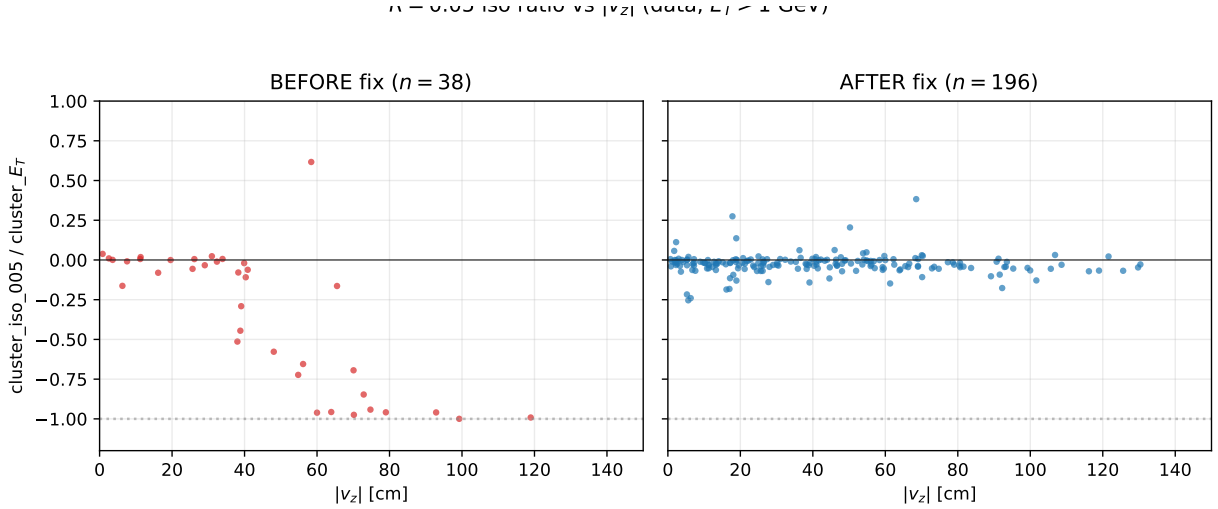


Figure 2: `cluster_iso_005` / `cluster_ET` vs.  $|v_z|$  before (left, red) and after (right, blue) the fix. The pre-fix trend towards  $-1$  at large  $|v_z|$  is the  $\sim |v_z|/R$  depth-mismatch signature; it is absent after the fix.

The pathological rate binned in  $|v_z|$  makes the transition sharp (Fig. 3). Before the fix the fraction of clusters with `iso_005`  $\approx -E_T$  goes from 0 % below  $|v_z| = 40$  cm to 100 % above 80 cm; after the fix it is consistent with zero everywhere.

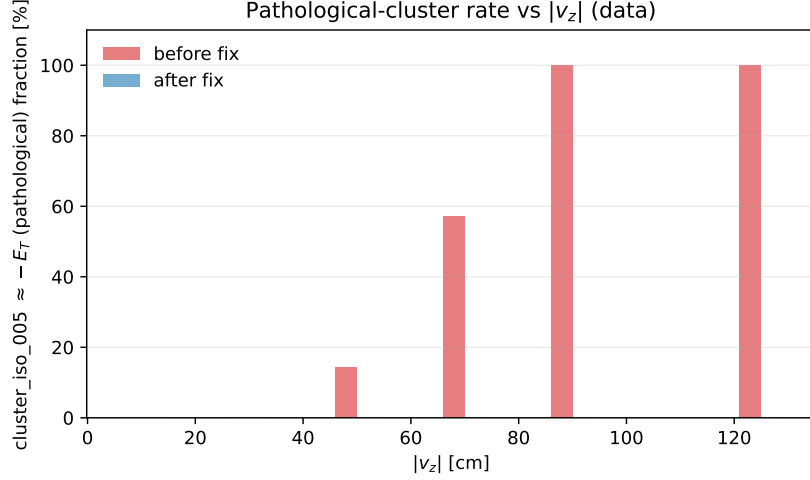


Figure 3: Fraction of clusters with `cluster_iso_005`  $\in [-1.1, -0.9] E_T$  (“pathological”), binned in  $|v_z|$ .

## 1.2 Monotonicity cascade

A photon cluster’s isolation sum should be a non-decreasing function of cone radius. Plotting  $\text{iso}(R) / \text{cluster\_}E_T$  for each cluster as a line in  $R$ , the pre-fix sample shows many lines that start at  $-1$  (cone at  $R = 0.05$  captures nothing) and climb back to  $0$  as  $R$  grows enough to re-contain the cluster – a textbook mis-centering signature. Post-fix the ensemble is monotone and small, as expected (Fig. 4).

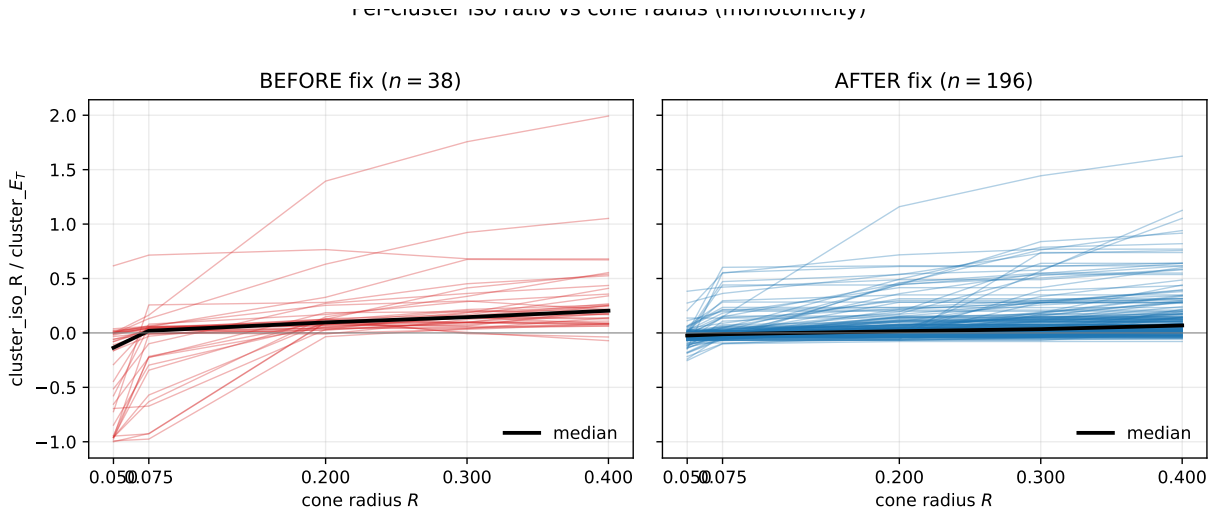


Figure 4: Per-cluster  $\text{iso}(R) / \text{cluster\_}E_T$  vs. cone radius. Black line: median over all clusters. Before fix (left): many clusters have  $\text{iso}(R = 0.05) \ll 0$ , walking back to zero as  $R$  grows past the mis-centering offset. After fix (right): monotone and bounded, as required.

## 2 Root cause

**Cone axis.** Line 1180 of `CaloAna24.cc` set `eta = RawClusterUtility::GetPseudorapidity(*recoCluster, v` which projects `recoCluster->get_position()` through `vertex = (0, 0, m_vertex)`. The sPHENIX cluster COG already has a shower-depth correction applied (the position sits at the energy-weighted *shower* depth,  $R_{\text{cluster}} \approx 95$  cm for a photon).

**Constituent frame.** In `calculateET` (tower iso) and `calculateET_topo_6cones` (topo iso) each constituent's  $\eta$  is obtained from `RawTowerGeom::get_center_*` (tower geometric center,  $R_{\text{tow}} \approx 102$  cm) or from the topo-cluster's energy-weighted cell centroid (also at tower mid-depth, and further out if HCal cells contribute).

**Projected offset.** For a photon at physics  $\eta_0$  and a vertex at  $v_z$ , the two paths project to

$$\eta_{\text{axis}} - \eta_{\text{const}} \simeq v_z \left( \frac{1}{R_{\text{cluster}}} - \frac{1}{R_{\text{tow}}} \right) \cosh \eta_0,$$

which for  $R_{\text{cluster}} = 95$  cm,  $R_{\text{tow}} = 102$  cm,  $v_z = 80$  cm,  $\eta_0 = 0.3$  gives  $\Delta\eta \approx 0.05$  — exactly at the edge of  $R = 0.05$  and the threshold where the cluster's own central tower falls outside the cone. For  $v_z = 100$  cm the offset grows to  $\approx 0.07$ , producing a clean 100 % miss rate at  $R = 0.05$ .

Ownership exclusion was not the primary cause: the pathology affects the topo-cluster path as well (which does not use identity-based self-exclusion and has no per-tower threshold to toggle), with the same  $|v_z|$ -dependence. That rules out the `isGood` filter, the tower-container choice, and the `-cluster_ET` scalar subtraction as explanations on their own.

### 3 Fix

Three independent changes in `CaloAna24.cc`, all on the same code path:

1. **CoG-tower axis.** Before the iso block, look up the central-tower geometry via `RawTowerDefs::encode_tow` and project through  $v_z$  using `getTowerEta(cog_tg, 0, 0, vertexz)` and `cog_tg->get_phi()`. These values, `iso_axis_eta` and `iso_axis_phi`, are passed to every `calculateET(...)` and `calculateET_topo_6cones(...)` call. Cone axis and constituents now live on the same reference surface; the  $|v_z|$ -dependent offset cancels.
2. **Ownership-excluded variant.** A new helper `calculateET_excl(eta, phi, dR, layer, min_E, skip_` mirrors `calculateET` but skips towers whose `TowerInfoDefs::encode_emcal` key is in the skip set. The set is built from `recoCluster->get_towermap()` for each cluster. Six new branches `cluster_iso_excl_{005, 0075, 01, 02, 03, 04}` are written as  $\sum_{\text{layers}}$  without the scalar `-cluster_ET` subtraction.
3. **Unified 120 MeV tower threshold.** All manual full-calo iso branches (`cluster_iso_{005, 0075, 02, 03, 04}` and `cluster_iso_excl_*`) use `min_E = 0.12` GeV. The built-in `RawCluster::get_et_is` accessor (no threshold, cluster-COG axis) is no longer read; its previous consumers `cluster_iso_{02, 03, 04}` are replaced by manual sums. Per-layer 60/70/120 MeV families are unchanged.

The header (`CaloAna24.h`) gains `#include <set>`, the six `cluster_iso_excl_*` data members, and the `calculateET_excl` declaration.

## 4 Verification

### 4.1 Before/after on data

Table 1 summarizes the pathological rates binned in  $|v_z|$ . The pre-fix numbers come from a slintree produced by the committed HEAD before this change; the post-fix numbers come from re-running the same `anatreemaker` macro with the fix applied.

Table 1: Pathological-cluster fraction (`cluster_iso_005`  $\in [-1.1, -0.9] E_T$ ) in Run 24 data (run 47289, segment 0,  $E_T > 1$  GeV).

| $ v_z $ bin | $n_{\text{before}}$ | pathol before  | $n_{\text{after}}$ | pathol after |
|-------------|---------------------|----------------|--------------------|--------------|
| 0 – 40 cm   | 20                  | 0.0 %          | 103                | <b>0.0 %</b> |
| 40 – 80 cm  | 14                  | 35.7 %         | 65                 | <b>0.0 %</b> |
| 80+ cm      | 4                   | <b>100.0 %</b> | 28                 | <b>0.0 %</b> |
| <b>all</b>  | <b>38</b>           | <b>23.7 %</b>  | <b>196</b>         | <b>0.0 %</b> |

Per-cluster monotonicity is also restored: post-fix, 100 % of clusters satisfy  $\text{iso}_{005} \leq \text{iso}_{02} \leq \text{iso}_{04}$  (within a 1 MeV rounding slop), against frequent violations pre-fix.

## 4.2 New iso\_excl family

The `cluster_iso_excl_*` branches are now the preferred small- $R$  iso measure: the cone axis is on the correct frame, the cluster’s own towers are removed by identity, and there is no scalar- $E_T$  subtraction that could drive the value unphysically negative. Figure 5 shows the post-fix distributions in data: non-negative populations growing with cone radius, as expected for a pure neighbor-energy isolation.

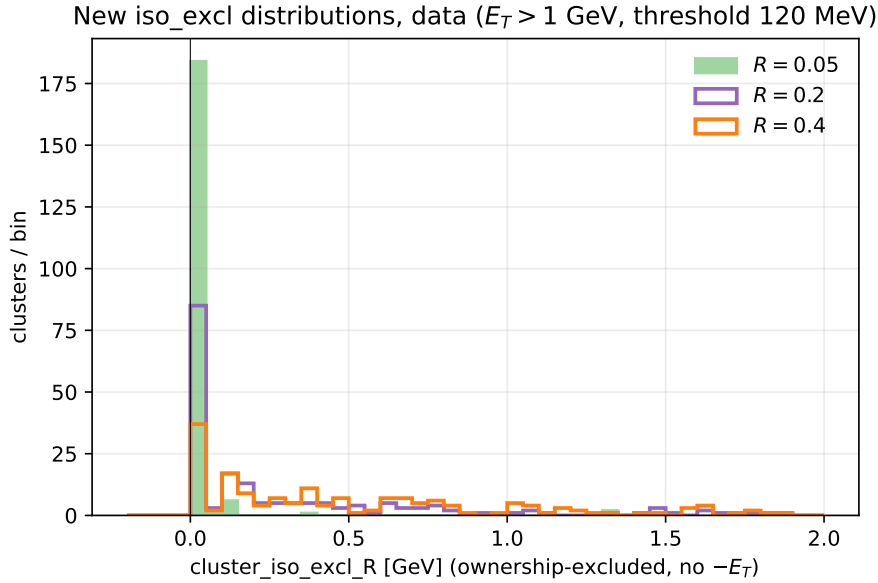


Figure 5: Post-fix `cluster_iso_excl_R` distributions at  $R = 0.05, 0.2, 0.4$  (data,  $E_T > 1$  GeV, 120 MeV tower threshold).

## 5 Downstream implications

- The nominal PPG12 isolation selection is on `cluster_iso_topo_04` with `reco_iso_max = reco_iso_max_b` (parametric cut in the efficiency configs). The topo path gets the same CoG-tower-axis fix; the  $|v_z|$ -dependent tail is removed. For low- $|v_z|$  clusters the nominal iso is essentially unchanged. For high- $|v_z|$  clusters, pre-fix values that had been artificially pushed toward  $-E_T$  now sit at the physical value, so those clusters migrate from noniso into iso of the ABCD regions (affects  $A, B, C, D$  populations and consequently purity and efficiency).
- `cluster_iso_{02, 03, 04}` now come from the manual 120 MeV-threshold path rather than the built-in `get_et_iso` (no threshold, cluster-COG axis). Downstream code that reads

these branches (e.g. `efficiencytool/*.C`, `FunWithxgboost/apply_BDT.C`, `BDTinput.C`) will see numerically different values even for low- $|v_z|$  clusters — the threshold change alone shifts the iso scale. The parametric iso cut (`reco_iso_max_b`, `reco_iso_max_s`) may need retuning on the new distribution.

- Production of fresh slimtrees is required before any downstream efficiency / purity / cross-section re-run. The fix is committed on `main` at `89941f1` (`anatreemaker`).

## Files

- `anatreemaker/source/CaloAna24.cc` — cone-axis replacement (L1207–1246), per-layer 120 MeV sums for  $R = 0.1, 0.2, 0.4$  (L1291–1302), `_excl` callers (L1303–1308), `iso_02/03/04` manual fills (L1862–1864), `iso_excl` branch writes (L1884–1890), `calculateET_excl` implementation (L2447–2518).
- `anatreemaker/source/CaloAna24.h` — `<set>` include, six `cluster_iso_excl_*` data members, `calculateET_excl` declaration.
- `reports/figures/iso_cone_fix/make_plots.py` — diagnostic script that reproduces every figure in this note from the two slimtreen test outputs.